Sultan Qaboos University
Department of Computer Science
COMP4609: Deep Learning Fundamentals, Spring 2025
Course Project

Task description

Your task in this project is to develop (train and evaluate) a deep learning neural network model that can perform recognition of alphanumeric sequence from car license plates images in Oman.

Project Dataset

The dataset contains image crops (cut out sections) of regions of car plate used in Oman and the corresponding sequence of numbers and letters (alphanumeric) as shown in the figure below.

The input to your neural network model (X) is an image and the output must be the sequence of alphanumeric code (y) without the word عمان, for example, the image in the top right corner contain the sequence "87796BB" which should be predicted by the neural network. Note that not all the images have the same dimensions/resolution. I recommend using consistent dimensions of 50x150 (using appropriate image resize function) but you are not strictly required to use this dimension depending on your choice of neural network architecture or design. You will get the training (300 images) and validation (101) datasets, which you can use for model development. <u>Remember that it is very important/wise to keep the validation set only for evaluation purposes during development and never use it for training the model to prevent biased results.</u>

Deep Learning Model

You must demonstrate that you have fully explored and experimented with a specific deep learning model architecture and that you have thoroughly tested/evaluated different strategies to improve it's performance. In particular, the following are some points (but not all) that you need to consider when you develop your model/strategy:

- Preprocessing Steps:
  - Image resizing and grayscale-color transformation.
  - Image Normalization (pixel values scaling)
  - Image augmentation (rotation, noise, shifting)
  - Image segmentations techniques (threshold-based, edge-based, deep-learning based…etc)
- Justification of preprocessing choices.

- Neural network desgin/archetictures:
  - Feature Extraction
  - (CNN) Convolutional layers (filter size, kernel size, activation functions).
  - Pooling layers (max/average pooling).
  - Sequence Modeling layers (RNN/LSTM/GRU/Transformer).
  - Number of layers, hidden units, type of activation functions.
- The Output Layer and Loss functions
  - Softmax activation for classification and/or CTC loss for sequence handling.
- Justification for model design choices.

- Optimization Algorithm: Adam, RMSprop, SGD, …etc.
- Hyperparameters:
  - Learning rate tuning.
  - Batch size selection.
  - Number of epochs.
- Regularization Techniques:
  - E.g., Dropout and batch normalization.
- Training/validation curves & analysis.

The above list is just suggestions on what needs to be thought about when you desgin your neural network model. Since this is a project you are required to do you own research/reading to device the best strategy for this problem and the skies are the limit ☺ as long as you clearly jusfiy based on readings.

Baseline System

The baseline is extremely important to compare any meaningful improvements in the task. I will be releasing a baseline model for you, however, for your model development you **<u>must</u>** demonstarte initial results you obtain from the initial design of your neural network. Based on this baseline you show what you have done to improve the initial results with clear justifications.

Task evaluation metrics

Your submissions will be evaluated on the held-out test set using Character Error Rate (CER) and accuracy metric. CER metric (lower is better) will be used as the primary metric (why?) for **ranking** the different submissions. The evaluation code will be made available to each team to utilize it for measuring their performance on the `validation` data during development. In general, the performance on the `validation` data should be used as a proxy for expected performance on the `test` data. The better the system is on validation the better it **<u>should</u>** score on the test set although this is not **<u>guaranteed</u>**. You must prepare **<u>at least</u>** 3 submissions for final submission and evlaution (more is allowed) on the test set. I will be realeasing more guidelines later about the format of the submissions.

Project Rules

1. There are **tons** of resources and ready codes/packages/models available on the internet for this recognition task. While it is usually good to reuse code, and gain state-of-the-art performance; our aim here is essentially to learn and be novel/creative. If you decide to reuse an existing code or system, then you must be able to demonstrate that you understand the code clearly and be able to modify it for your own desing/strategy. You MUST be able

to demosntrate that you understand the design of the full model (including being able to build it from scratch) layer-by-layer even you if use existing librarie/code to generate the model. **<u>Using existing code or system from the internet as black box with end-to-end to only generate the required output and submit blindly as your contribution is strictly not allowed. If you are in doubt then ask.</u>**

2. Each team of two students must propose **<u>three strategies</u>** to implement for the task. This means that each team must make one file submission containing the predictions from each proposed strategy (3 file submissions). You may of course adopt an approach published in **<u>a prior research</u>** work with the condition that you will be able to demonstrate you understand the algorithms/methods/model used.
3. Any submission which rely on only training/fine-tuning deep learning model as blackbox i.e., feeding it X (input text) and y (expected output); will not be acceptable. YOU NEED to demonstrate that you have PROPOSED and EXPLORED different approaches/stratgeies and analyzed the errors that it makes rigorously and then tried different techniques to improve its performance. **<u>Any submission relying on use of black-boxes will be heavily penalized!</u>**
4. The idea of baseline and iterative improvements is that you start with an approach figure out it's weakness and then propose some modifications to improve it's performance. This means that the other strategy should be related to some extent to your analysis of the first strategy and its weaknesses. Of course, you may also include in your presentation/report any other attempts which failed to improve scores (which can be counted as an effort)
5. Avoid the **<u>terrible error</u>** which rely on the fact that the CSV (expected sequence output) will be given to the model as input during testing/evaluation WHICH IS NOT!! The only thing your model will receive during evaluation is the raw image files and nothing else (i.e. NO labels are available). Correct output is used for **evaluation** only during testing time.
6. Note that scoring the highest and being ranked best (1st position) among the final submissions is **<u>an excellent aim</u>** (maybe some prizes will be given ;), however, it is certainly not going to affect your grade or mark. In fact, the focus is on the novelty of the approaches being used (methods which were never tried before) regardless of the final rank. Your contribution could be ranked last in the list and yet maybe the most novel of all submissions. **<u>In short, rankings may not correlate with grades!</u>**

7. As usual any use of materials/codes **<u>must be clearly</u>** attributed and indicated in your submissions.

Project evaluation

Your contribution will be roughly evaluated based on **completeness** (you have completed all project requirements) and **<u>sophistication/intricacy/creativity</u>**. The second criterion roughly means how sophisticated the approach and methods you proposed and used. It also, roughly means that you successfully demonstrated creative thinking by doing error analysis and demostrating choices/stratgeis that were selected/proposed based on early results. The following table also list some of the points that you need to consider for your development and final submission:

Category	Excellent	Good	Satisfactory	week

Data Preprocessing	Dataset well-prepared, with clear and justified preprocessing steps	preprocessing steps applied, but justification is weak or inconsistent.	Basic preprocessing applied	Poorly prepared dataset, missing major preprocessing steps, or improper justification.
Model Architecture	Well-structured/designed NN model or CNN-RNN/Transformer model with strong justification for design choices.	Model is functional but lacks full optimization or justification.	Basic model implemented but with incorrect/missing layers or weak justification.	Poor model architecture, ineffective design, or major errors in implementation.
Training Strategy & Hyperparameters	Well-optimized hyperparameters with documented tuning (learning rate, batch size, epochs). Training/validation curves analyzed thoroughly.	Basic tuning attempted, but lacks detailed analysis.	Poorly chosen hyperparameters with minimal improvement or tuning effort.	Default values used with no optimization effort or training issues ignored.

Project deliverables and Submissions

Submit through elearn website the following (exact deadlines will be indicated in the elearn website):

- Project Progress report and code in Jupyter Notebook (week 7) – 5%
- Project final report and code -Jupyter Notebook- and presentation (week 14) – 15%